

Effect of a single high-fat meal on endothelial function in healthy subjects.

Although there is a well-established relation between serum cholesterol and coronary artery disease risk, individual and national variations in this association suggest that other factors are involved in atherogenesis. High-fat diet associated triglyceride-rich lipoproteins have also been suggested to be atherogenic. To assess the direct effect of postprandial triglyceride-rich lipoproteins on endothelial function, an early factor in atherogenesis--10 healthy, normocholesterolemic volunteers--were studied before and for 6 hours after single isocaloric high- and low-fat meals (900 calorie; 50 and 0 g fat, respectively). Endothelial function, in the form of flow-mediated vasoactivity, was assessed in the brachial artery using 7.5-MHz ultrasound as percent arterial diameter change 1 minute after 5 minutes of upper-arm arterial occlusion. Serum lipoproteins and glucose were determined before eating and 2 and 4 hours postprandially. Serum triglycerides increased from 94 $\pm$ 55 mg/dl preprandially to 147 $\pm$ 80 mg/dl 2 hours after the high-fat meal ($p = 0.05$). Flow-dependent vasoactivity decreased from 21 $\pm$ 5% preprandially to 11 $\pm$ 4%, 11 $\pm$ 6%, and 10 $\pm$ 3% at 2, 3, and 4 hours after the high-fat meal, respectively (all $p < 0.05$ compared with low-fat meal data). No changes in lipoproteins or flow-mediated vasoactivity were observed after the low-fat meal. Fasting low-density lipoprotein cholesterol correlated inversely ($r = -0.47$, $p = 0.04$) with preprandial flow-mediated vasoactivity, but triglyceride level did not. Mean change in postprandial flow-mediated vasoactivity at 2, 3, and 4 hours correlated with change in 2-hour serum triglycerides ($r = -0.51$, $p = 0.02$). These results demonstrate that a single high-fat meal transiently impairs endothelial function. These findings identify a potential process by which a high-fat diet may be atherogenic independent of induced changes in cholesterol.